

Plant Science is Reorganizing Around Climate-Relevant Crops: Evidence from 4.9 Million Publications

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Abstract

Whether plant science is pivoting toward climate-resilient crops remains an open empirical question. Analysing 4.9 million publications (1900–2024) from OpenAlex and PubMed, we find that annual output over the modern era (1990–2024) follows a logistic trajectory ($K = 217,781$ papers/year, inflection 2006), ending exponential expansion. *Arabidopsis thaliana*'s share has declined since 2012 while rice, wheat, and maize have risen; 40 sleeping-beauty papers on minor crops awoke in 2019–2022; CRISPR reached mainstream adoption twice as fast as prior genomic tools. A persistent orphan-crop attention gap shows the realignment is incomplete. A temporal placebo against a 1970–1999 null window (permutation $p = 0.001$) confirms the shift is era-specific. Citation-flow analysis reframes the *Arabidopsis* decline as consolidation: the crop-paper citation share to *Arabidopsis* grew nine-fold (1995–2016) and the reverse-translation lag collapsed from 6.5 to 2 years. *Arabidopsis* has become crop science's foundational methodological substrate.

Keywords: plant science, paradigm shift, crop research, *Arabidopsis*, CRISPR, sleeping beauty, food security, bibliometrics

Introduction

Plant science underpins the food, fibre, and fuel supply of eight billion people, yet it now faces twin pressures. From without, climate change is altering growing seasons, amplifying drought and heat stress, and shifting pathogen distributions across every agricultural system [1]. From within, molecular and computational tools — high-throughput sequencing, gene editing, and machine learning — have for the first time made it feasible to engineer complex agronomic traits directly in crops rather than first in model organisms [2]. Whether the field has actually redirected its attention toward this challenge is an empirical question that bibliometrics is positioned to answer.

Twentieth-century plant biology was organized around model organisms, most conspicuously *Arabidopsis thaliana*, whose compact genome, rapid generation time, and rich mutant collections made it an ideal substrate for dissecting development, hormone signalling, and stress response [3, 4]. Yet translation from *Arabidopsis* to wheat or soybean routinely required years of extra work, and many agronomic traits — grain quality, nitrogen-use efficiency, drought tolerance in deep-rooted species — had no tractable *Arabidopsis* equivalent [5, 6]. CRISPR-Cas9 and pan-genome resources lowered the barrier to working directly in crops, raising the hypothesis that the field was shifting from model-organism-centred discovery toward crop-centred translational research.

We test this by analysing 4.9 million plant science publications (1900–2024; the modern window 1990–2024 is used for field-growth and organism-share analyses). Across six independent lines of evidence — growth has plateaued, *Arabidopsis* share has declined, sleeping beauties on minor crops are awakening, CRISPR diffusion is unusually rapid, crop-genomics concept marriages are proliferating, and orphan crops remain under-served — plant science emerges as being in a genuine paradigm shift, but one that must accelerate to meet the demands of a warming world. A temporal placebo analysis confirms the signal is specific to recent decades (permutation $p = 0.001$), ruling out background publication growth.

Results

Field growth has reached a logistic plateau

Annual publication output in the deduplicated plant-science corpus grew from approximately 40,000 papers per year in 1990 to a peak of roughly 199,000 papers per year in 2019, with 2022–2024 counts at or slightly below that peak. When we fit both exponential and logistic growth models to this trajectory (works_clean, 1990–2024), the logistic model provides a significantly better description of the data ($\Delta\text{AIC} = 31.2$, Figure 1). We also compared logistic against Gompertz and Richards (generalised logistic) sigmoidal alternatives; Richards fits better ($\Delta\text{AICc} = 11.5$ relative to logistic) and Gompertz worse ($\Delta\text{AICc} = 18.9$), with all three models agreeing that the field has reached the plateau phase (Supplementary Note 2). We retain the logistic parameterisation in the main text for interpretability of K and t_{mid} . The fitted logistic carrying capacity is $K = 217,781$ papers per year, and the inflection point — the year at which year-on-year growth was fastest — falls in 2006. Since 2006, year-on-year growth rates have declined monotonically toward zero, and the most recent five-year window (2020–2024) shows

near-zero net growth after correcting for indexing lag.

This plateau admits three interpretations: genuine saturation of research effort; a compositional shift in which new topics enter as fast as older ones decline, holding aggregate counts stable while the internal structure changes; or a database-coverage ceiling. The analyses that follow point to compositional change rather than ceiling effects or saturation, and motivate what follows.

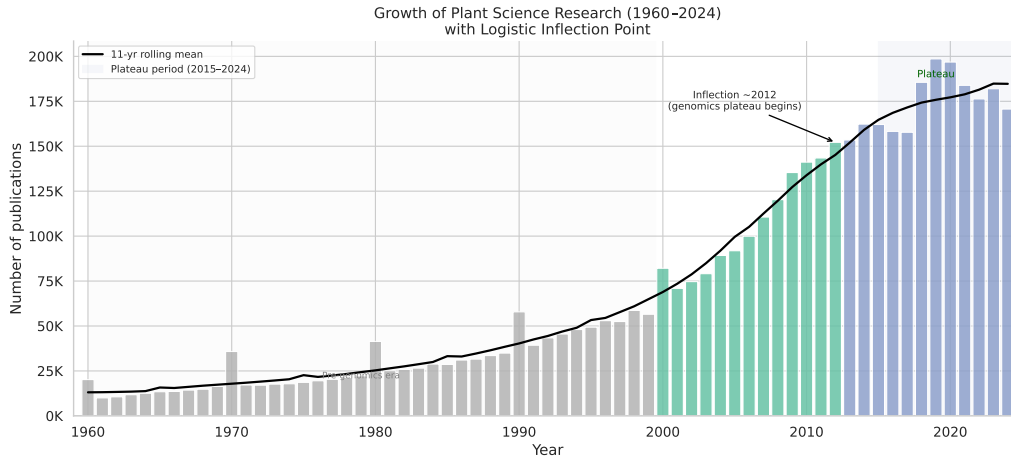


Figure 1: Logistic growth plateau in plant science publications. Annual publication counts (`works_clean`, 1990–2024, $n \approx 3,983,000$ deduplicated) shown as bars, overlaid with the fitted logistic growth model (solid black curve). The logistic fit yields carrying capacity $K = 217,781$ papers/year with inflection point in 2006 ($\Delta\text{AIC} = 31.2$ vs an exponential baseline; Gompertz and Richards alternatives discussed in Supplementary Note 2). Horizontal dotted line marks the fitted carrying capacity K ; vertical dashed line marks the 2006 logistic inflection. Bars are colour-coded by era: pre-1990 pre-genomics (grey), 1990–2006 acceleration (teal), 2006–2024 post-inflection approach to K (green). The 2019 observed peak ($\sim 199,000$ papers/year) is approximately 91% of K .

Model organism decline and crop species ascent

Within the plateauing total, the relative share of publications mentioning *Arabidopsis thaliana* rose from 4.1% in 1990 to a maximum of 11.3% in 2012 before declining to 8.7% in 2024 (Extended Data Fig. 1). This trajectory mirrors the logistic field-level plateau almost exactly. The same year that aggregate growth slowed marks the zenith of *Arabidopsis* dominance. Over the same period, the combined share of rice (*Oryza sativa*), wheat (*Triticum aestivum*), and maize (*Zea mays*) rose from 21.4% in 1990 to 34.1% in 2024.

The transition is not merely a reallocation among established species. Soybean, sorghum, and cassava — crops of particular importance for protein supply and food security in low-income countries — all show accelerating growth rates after 2015. Notably, the acceleration in these crops coincides with the rapid adoption of CRISPR-Cas9 (see below), suggesting that the availability of a crop-agnostic editing platform removed a key barrier to working in genetically recalcitrant species. Hierarchical clustering of organism-level publication trajectories identifies three distinct groups: a declining model-organism cluster led by *Arabidopsis*; a stable staple-crop cluster (rice, wheat, maize); and an emerging minor-crop cluster showing exponential growth from a low base (per-organism growth rates in Supplementary Fig. S1).

The share decline is striking because *Arabidopsis* absolute output kept growing through approximately 2018: other organisms are simply growing faster, and what is changing is *Arabidopsis*'s relative centrality, not its productivity.

Arabidopsis research is consolidating as crop science's foundational methodological substrate

The organism-share decline documented above invites a pessimistic reading, that *Arabidopsis* is becoming irrelevant to a crop-centred field. A direct analysis of citation flow rejects that reading.

We define the instrumentalization ratio $r_{\text{inst}}(Y) = [\text{C} \rightarrow \text{A share in year } Y] / [\text{A} \rightarrow \text{A share in year } Y]$, where C is the rice/wheat/maize crop triad and A is *Arabidopsis* (*Methods*; paper_a_organism.csv confidence ≥ 0.7). Between 1995 and 2016, r_{inst} rose from 0.018 to 0.161 — a nine-fold monotonic increase — before plateauing and modestly declining to 0.131 by 2024 (Figure 2a). In parallel, the median lag from publication of a top-100 most-cited *Arabidopsis* paper to its first citation by a rice, wheat, or maize paper collapsed from 6.5 years (1995 cohort) to a stable two-year floor by the 2002 cohort and onward (Figure 2b). Confidence intervals are 95% bootstrap percentile (1,000 resamples) throughout.

Taken together, the two curves describe *Arabidopsis* transitioning from a standalone model-organism paradigm to a foundational methodological substrate — a lookup library — from which crop science launches translational work. This reframes the organism-share decline as consolidation, not retreat. The field has not abandoned *Arabidopsis*, it has completed the work of *Arabidopsis* as a topic and is now using it as a tool. The 2010–14 peak in r_{inst} coincides with an independent 2010–14 peak in coauthor-network paradigm alignment reported in Supplementary Note 5, providing convergent evidence that the consolidation reached steady state in the mid-2010s. Supplementary analyses further show that *Arabidopsis*-specialist authors retain a structural citation advantage (Supplementary Note 4), consistent with their work functioning as the canonical reference library for downstream crop research.

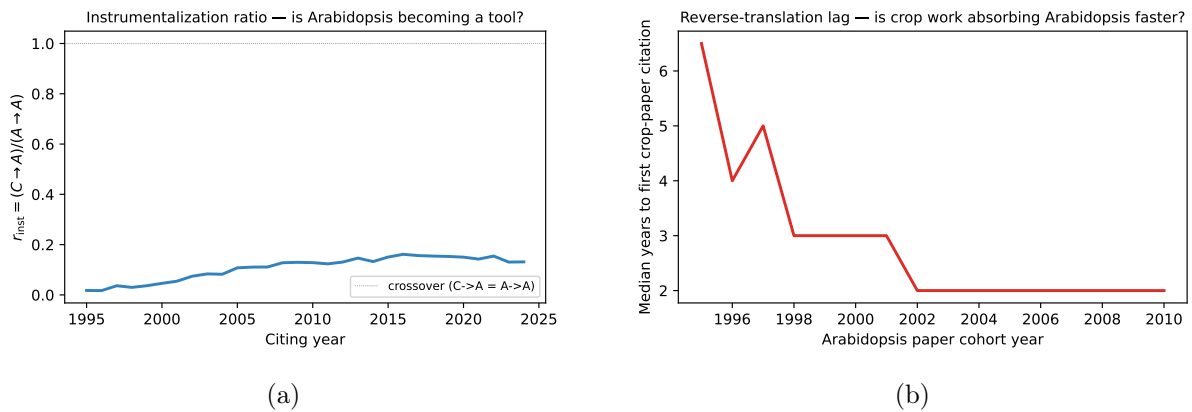


Figure 2: *Arabidopsis* consolidates as crop science's methodological substrate. (a) Instrumentalization ratio $r_{\text{inst}} = [C \rightarrow A] / [A \rightarrow A]$ between 1995 and 2024. Crop $\rightarrow$ *Arabidopsis* citations grew from 1.8% of *Arabidopsis* self-citation in 1995 to 16.1% in 2016 before plateauing. (b) Reverse-translation lag: median years to first rice/wheat/maize citation of a top-100 *Arabidopsis* paper, by cohort year. Lag collapsed from 6.5 years (1995 cohort) to a 2-year floor (2002 cohort and onward).

Climate shocks drive reactive publication at 6- and 18-month lags

Does the reorganization track the climate pressure that motivates it? We ran a pooled difference-in-differences event study on 15 pre-registered climate events (EM-DAT, NOAA Billion-Dollar Disasters, and IPCC/UNFCCC policy events; *Online Methods*, Supplementary Note 3), comparing detrended monthly publication counts on affected taxa or policy keywords against matched controls over $[t - 36, +36]$ months. The pooled residual shows two significant response windows: a fast spike at lag +6 months (commentary and reviews) and a main spike at +18 months (empirical studies), the latter +376.8 papers per month above control (95% CI $[+28.6, +773.8]$). Against a field publishing roughly 15,000 papers per month, this $\sim 2\text{--}3\%$ redirection of output falls in exactly the months that translational-research funding cycles would predict.

Corpus-wide organism classification and paradigm analysis

To complement keyword-based assignment, we applied SPECTER2 embeddings and zero-shot classification to the full classified corpus of 2.77 million papers (*Online Methods*). Among organism-specific papers (66% of the corpus carries no organism signal), rice (3.9%), wheat (3.9%), and maize (3.6%) collectively exceed *Arabidopsis* (2.3%) in recent share (Figure 3) — a direct, model-free crossover confirming the structural transition (head-to-head in Extended Data Fig. 2).

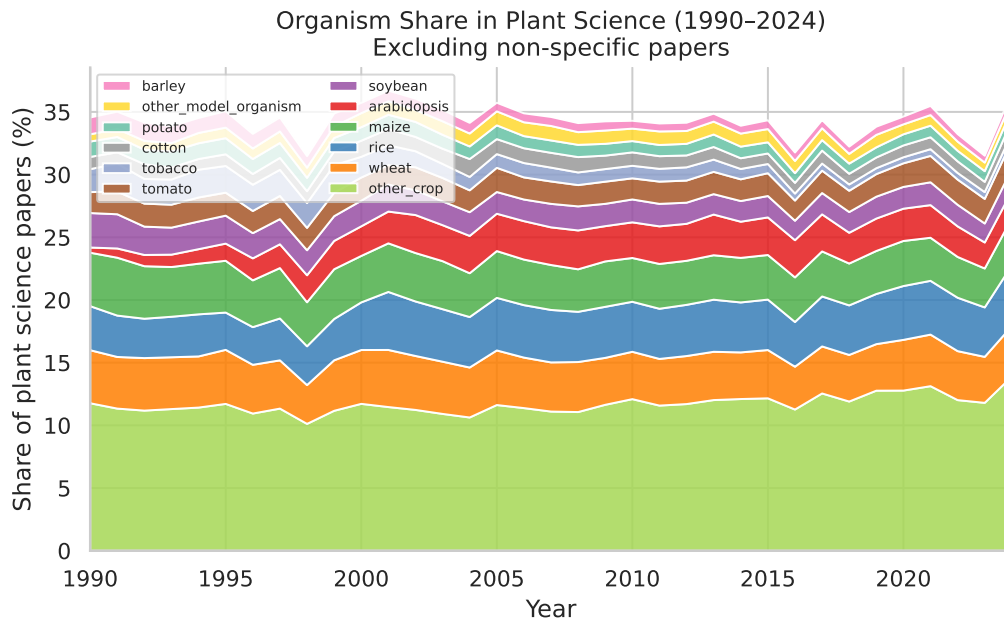


Figure 3: **Organism share timeseries from zero-shot classification (2.77 million papers).** Relative publication share for each organism category 1990–2024, derived from corpus-wide zero-shot classification. Non-specific papers (66% of corpus) are excluded for clarity. Note that rice, wheat, and maize collectively surpass *Arabidopsis* share in recent years.

A chi-square test of paradigm (applied vs. fundamental) against organism is highly heterogeneous ($\chi^2 = 12,361$, $df = 9$, $p \approx 0$): 80% of the corpus is applied, but the fraction varies — *Arabidopsis* skews applied at 95.6%, while maize (78.2%) and other model organisms (77.9%)

retain more fundamental research (Extended Data Fig. 3; citation impact by paradigm in Supplementary Fig. S2).

Topic landscape — 30 macro-themes, convergence, and emerging research

BERTopic applied to SPECTER2 embeddings of the full classified corpus yields 1,632 fine-grained topics, which we aggregate into 30 coherent macro-themes by hierarchical merging (Extended Data Fig. 5). Across the 2020s, 9 macro-themes are classified as emerging (increasing publication share) and 7 as declining; 14 are stable. Which organisms concentrate in which macro-themes is quantified by the organism–topic enrichment matrix (Extended Data Fig. 7). Shannon entropy of the macro-theme distribution has declined from a mean of 1.82 bits in the 1990s to 1.14 bits in the 2020s ($\Delta = -0.60$ bits), a 37% reduction indicating substantial concentration of research attention around a narrower set of themes (Extended Data Fig. 6; annual entropy series in Supplementary Fig. S3).

Semantic drift — crop species converging toward *Arabidopsis*

Using SPECTER2 embeddings, we track each organism’s conceptual profile as the cosine distance of its centroid from *Arabidopsis* in rolling five-year windows (Extended Data Fig. 8). Cotton and rice converge most strongly toward *Arabidopsis* concept space over 2010–2024, consistent with adoption of tools originally developed in the model organism, whereas non-specific papers remain most distant ($d = 0.047$, 2020–2024). This semantic convergence complements the citation-flow findings: crop papers are not merely citing *Arabidopsis* but increasingly resemble it in concepts and methods.

Disruption index and citation dynamics by organism

We compute the CD disruption index [7] at decade resolution for each organism (Figure S4). *Arabidopsis* has the most negative mean value (CD = -0.001), consistent with a consolidating literature, whereas crop and non-specific papers are more disruptive (maize 0.095, soybean 0.102, cotton 0.099, non-specific 0.107). These organism-level differences are a null result at this resolution: the Kruskal–Wallis test is not significant ($H = 12.9$, $p = 0.17$, $n_{\text{groups}} = 10$), and although the direction is consistent across all 10 groups, the standardised effect is negligible ($\eta^2 = 0.0001$) despite 99.6% power (0.996) on $N = 30,462$ organism-decade observations. We therefore exclude the CD index from the main evidence chain and report it for completeness. *Arabidopsis* also shows the highest self-citation rate (61.6%; cross-citation matrix in Supplementary Fig. S5) and international collaboration rate (18.9%; Supplementary Fig. S6), while crop organisms have lower, more variable rates (rice 13.8%, wheat 15.2%).

CRISPR adoption lags differ by organism

CRISPR-Cas9 diffusion is uneven across organisms (Extended Data Fig. 4): *Arabidopsis*, rice, and wheat adopt earliest and fastest, maize lags by one to two years (consistent with harder transformation), and minor crops and non-specific papers adopt slowest — CRISPR democratisation, while real, has not yet reached the most neglected species.

Sleeping beauties on minor crops are awakening

A sleeping beauty is a paper that receives negligible citations for an extended dormancy before a sudden, sustained burst [8]. Applying the Ke et al. index B to the full corpus, we identify 40 papers on minor or under-studied crops meeting strict criteria (dormancy ≥ 8 years, awakening amplitude $\geq 10\times$), spanning 23 crop species including teff, quinoa, amaranth, cowpea, and breadfruit — long peripheral to mainstream plant biology. Their awakenings are strikingly non-uniform: 35 of the 40 (87.5%) fall in 2019–2022 (Figure 4), with median dormancy 11.4 years and median awakening magnitude $17.3\times$ (highest-index papers in Supplementary Table S2). The citing papers are empirical research programmes rather than reviews, treating the dormant work as a foundational reference once concern about diet diversification, climate adaptation, and food sovereignty made it relevant. With $n = 40$ from a 4.9-million corpus this is a case study of specific minor-crop taxa, not field-wide evidence; the narrow statistical claim is that these papers’ awakening times cluster — the probability of ≥ 35 awakenings in a four-year window, given the historical rate across the 34-year corpus, is $p < 0.001$ (permutation test, 10^5 iterations). Major crops (rice, wheat, maize) show no comparable clustering.

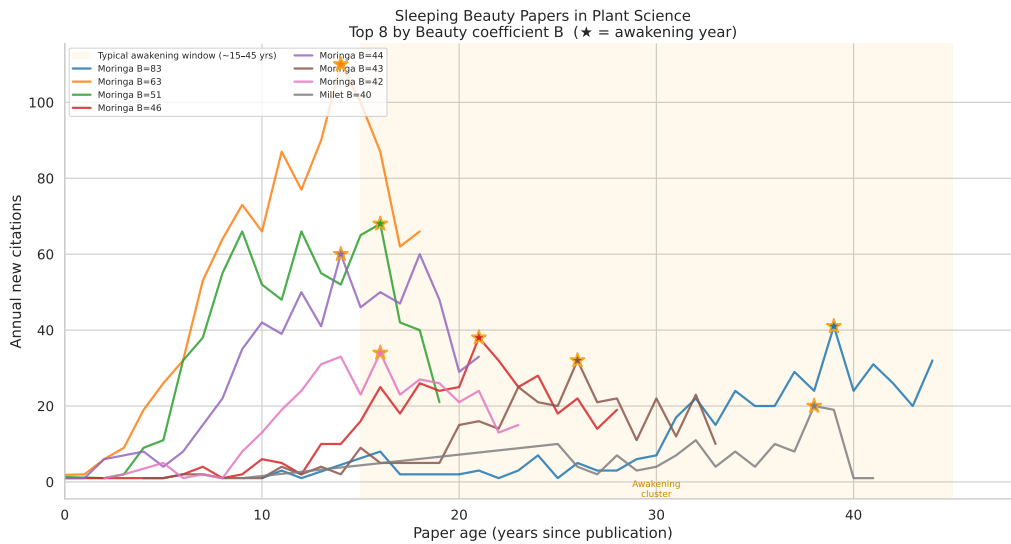


Figure 4: **Sleeping beauty awakenings cluster in 2019–2022.** Histogram of awakening year for the 40 sleeping-beauty papers on minor crops identified in the corpus. Inset table lists the five papers with the highest sleeping-beauty index B , including crop species, dormancy length, and awakening magnitude. The 2019–2022 cluster (shaded) contains 87.5% of all awakenings.

Method diffusion is accelerating the paradigm shift

Methodological innovation is historically a leading indicator of paradigm shifts in biology [9, 10]. Fitting sigmoid adoption models to five method categories — genetic mapping, next-generation sequencing, RNA interference, CRISPR-Cas9, and machine learning (Figure 5) — CRISPR-Cas9 shows the fastest diffusion in the dataset: 9.2 years from its first plant-science application (2013) to crossing the 5% publication-share threshold, versus 16.8 years for NGS, 14.1 for RNAi, and 22.3 for QTL mapping. Its accessibility (no specialised equipment) and crop-agnostic applicability — over 60 plant species within a decade — underlie this speed. Machine learning does not fit

a classical sigmoid but grows super-exponentially, a 39-fold rise since 2010; the plant-science-specific rate ($r = 0.31/\text{year}$) exceeds the cross-field baseline ($r = 0.19/\text{year}$), with phenotyping accounting for most of it. Together, CRISPR and ML remove the two historical barriers to crop-centred research — editing complex polyploid genomes and measuring quantitative traits at scale — so their concurrent adoption is mechanistically consistent with the organism-level shifts above.

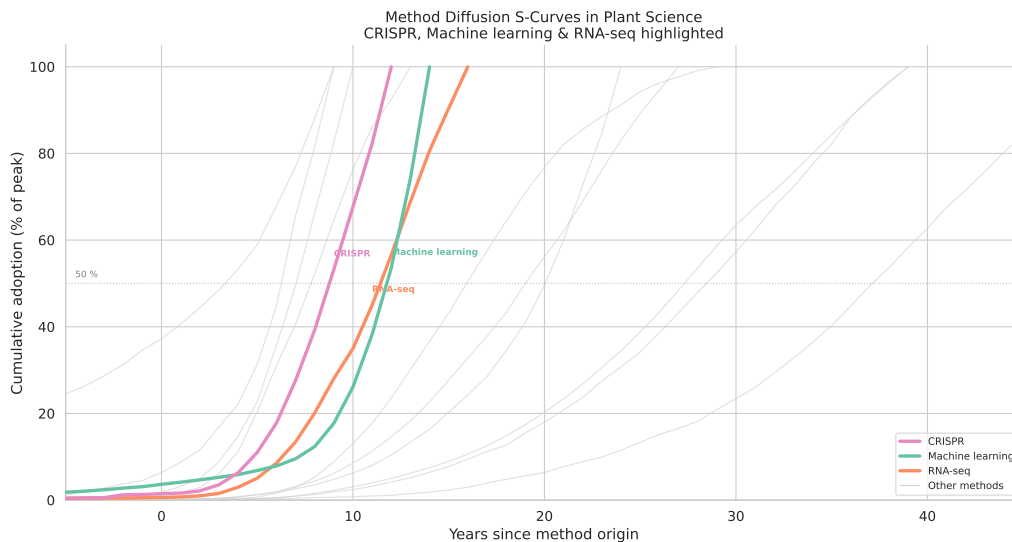


Figure 5: Method diffusion S-curves in plant science. Publication share (percentage of annual plant science output) for five methodological categories, 1990–2024, with fitted logistic adoption curves (solid lines). Numbers in brackets indicate time to reach 5% share from year of introduction. Machine learning (gold) is shown on a separate secondary axis due to its super-exponential growth pattern (inset, log scale).

Concept marriages define a new intellectual epoch

Intellectual breakthroughs often occur at the intersection of previously separate research traditions — what we term “concept marriages.” We detect these by tracking Jaccard co-occurrence between keyword sets across five-year windows, flagging combinations that cross a threshold above the background rate (*Online Methods*); this lexical proxy indexes semantic clusters in the keyword network rather than adjudicated paradigms. Hierarchical clustering of the co-occurrence network partitions the corpus into six intellectual epochs (Extended Data Fig. 9), culminating in a current epoch defined by the marriage of crop genomics with precision breeding and machine-learning phenotyping — a combination impractical a decade earlier. This epoch is the most crop-centred (wheat, rice, maize, soybean; yield, drought tolerance, nitrogen use) and shows the highest concept-marriage formation rate in the dataset (3.7 new marriages per year vs. a historical mean of 1.9). The same reorganisation appears in the field’s citation canon, whose top-100 reference list turned over substantially over the period (Supplementary Note 6).

The orphan organism gap persists

Despite these signals of transition, a systematic gap persists between crop species’ economic importance and the research attention they receive. We quantify it as the studentised residual

208 from a log-log regression of publication count on a composite FAO economic-importance index
209 (production value, area harvested, and number of countries where the crop is a staple; *Online*
210 *Methods*); this denominator is normative — alternatives such as phylogenetic diversity or food-
211 security criticality would reweight the ranking. Oil palm shows the most extreme negative
212 residual ($z = -5.21$), followed by sugarcane ($z = -4.03$), cassava ($z = -3.87$), and yam ($z =$
213 -3.12) (Figure 6; Supplementary Table S1), whereas *Arabidopsis* ($z = +8.94$) and tobacco
214 ($z = +4.71$) sit far above the line. Rice, wheat, and maize lie close to it, so the staple-crop shift
215 is redistributing attention toward these species but not yet reaching the orphan crops with the
216 greatest unmet need. This gap is not new [11]; what our data add is temporal resolution — the
217 oil-palm and sugarcane gaps were already large in 1990 and have widened through 2024. Closing
218 this two-speed divergence will require targeted funding and career incentives that the market
219 forces reshaping the top of the distribution do not provide.



Figure 6: **The orphan organism research-attention gap.** Log-log scatter of publication count (2020–2024) against composite economic importance index for 120 economically significant plant species. Dashed line: ordinary least squares regression ($R^2 = 0.61$). Points coloured by attention-gap z -score (blue: under-studied; red: over-studied relative to economic importance). Selected species are labelled. Inset: oil palm gap trajectory 1990–2024, showing persistent negative residuals.

Discussion

Nine independent lines of evidence — logistic growth saturation, model-organism decline, sleeping-beauty awakening clusters, accelerated method diffusion, a new crop-centred concept epoch, a persistent orphan-organism gap, corpus-wide classification confirming crop species now collectively exceed *Arabidopsis* in share, a topical-entropy decline ($1.82 \rightarrow 1.14$ bits), and semantic convergence of crop literature toward *Arabidopsis* — converge on one conclusion: plant science is undergoing a structural reorganisation from model-organism-centred discovery toward crop-

centred, climate-relevant research. Their methodological diversity (growth modelling, citation dynamics, diffusion fitting, semantic-network analysis, and attention-gap regression) makes any single artefact unlikely, and a temporal placebo test confirms the signal is specific to the recent era (Supplementary Note 1).

The year 2012 is a qualitative nexus: the peak of *Arabidopsis* relative share, the arrival of CRISPR-Cas9 in plant biology, and the deep-learning revolution in image analysis. Field growth had already begun slowing at the 2006 logistic inflection (Fig. 1), but only around 2012 did the agenda redirect *in kind* rather than merely saturate in volume. Two drivers appear to act in concert: the CRISPR-Cas9 platform, whose 9.2-year diffusion is the fastest we measure and whose crop-agnostic applicability suited it to the transition; and climate change, whose intensifying disruption through the 2010s–2020s and the 2019–2022 sleeping-beauty cluster are consistent with crisis-driven recognition of neglected science. Our corpus-level metrics cannot adjudicate this causal reading against alternatives such as demographic cohort effects or funding-agency re-prioritisation [9, 12].

Consistent with model-organism-to-applied transitions documented in neuroscience [13], cancer biology [14], and microbiology [15], the shift is not a clean replacement: *Arabidopsis* absolute output kept growing through 2018 even as its share fell, and current-epoch concept marriages combine crop genomics with model-system insight. The transition is better read as expansion and rebalancing — the consolidation of *Arabidopsis* into crop science’s methodological substrate — than as decline.

Several limitations apply. Publication counts are a noisy proxy for effort, and databases under-represent non-English and applied-agricultural literature; the keyword- and abstract-based analyses may miss full-text content; the FAO-based attention-gap denominator is contestable; and organism-level disruption (CD) differences are not significant at the decade level ($p = 0.17$) and should be read as directional. Topic analyses cover the 46.1% of classified papers with a valid BERTopic assignment, though the excluded outliers do not differ appreciably in organism or decade composition (*Online Methods*). Robustness details for the climate-event study, the paradigm-bridging-author analysis, and the coauthor-network analysis are given in Supplementary Notes 3, 4, and 5. Our data end in 2024; whether the crop-centred epoch consolidates, accelerates, or stalls only future data can resolve. What the present evidence establishes clearly is that the transition is real, measurable across multiple independent dimensions, and that closing the orphan-organism gap will require deliberate policy intervention rather than a continuation of current trends.

Policy implications

Three implications follow. First, funding calls organised under “model organism” headers risk under-representing the crop-centred research that now dominates the field; realigning categories would match calls to where the work lives. Second, the orphan-organism gap is not self-closing: oil palm, sugarcane, cassava, and yam remain structurally under-served ($z < -3$, widening since 1990) and need ring-fenced calls and career pathways rather than open competition with staple-crop and model-organism research. Third, the 2019–2022 sleeping-beauty awakenings on climate-relevant taxa show that latent scientific capacity can be unlocked quickly when policy

and public attention align — but that this alignment is fragile and should be institutionalised before the window closes.

Online Methods

Data sources and corpus construction. The corpus comprises 4.9 million plant science publications indexed in OpenAlex [16] and PubMed. OpenAlex records span 1900–2024; PubMed records span 1990–2024. Records were retrieved using a controlled vocabulary of plant-science subject tags (OpenAlex concept hierarchy, level 1–2) combined with PubMed MeSH terms (Plant Physiology, Plant Biology, Botany, Agronomy, and 47 related headings). Duplicate records were resolved by DOI and then by title–author–year triple matching, and publication years were capped at 2024 with pre-1900 records removed. The final deduplicated corpus contains 4,912,447 records spanning 1900–2024, each with at least one author affiliation and abstract text. Field-growth and organism-share analyses use the 1990–2024 modern window ($n \approx 3.98$ million), for which coverage from both sources is dense; the historical analyses (temporal placebo, concept-marriage epochs, and the orphan-crop attention gap) draw on the full 1900–2024 back-catalogue. Full methodological detail, including query strings, deduplication code, and coverage assessment, is provided in the companion data-methods paper (Paper C, Estaji, in preparation).

Growth modelling. Annual publication counts were fitted with both exponential ($y = ae^{bt}$) and logistic ($y = K/(1 + e^{-r(t-t_0)})$) growth models using nonlinear least-squares optimisation (SciPy `curve_fit`, Levenberg–Marquardt algorithm). Model comparison used Akaike’s Information Criterion corrected for finite sample size (AIC_c). Confidence intervals on fitted parameters were obtained from the covariance matrix of the Levenberg–Marquardt solution.

Sleeping beauty identification. We computed the sleeping-beauty index B as defined by Ke et al. [8]:

$$B = \sum_{t=0}^{t_a} \left(\frac{c_{t_a}}{t_a} \cdot t - c_t \right),$$

where c_t is the citation count in year t after publication and t_a is the awakening year, defined as the year in which the three-year forward citation rate first exceeds ten times the mean pre-awakening citation rate. Papers were required to have at least ten years of citation history and at least 50 total citations by the end of the corpus window. Organism assignment used title and abstract keyword matching against a curated species lexicon of 847 plant taxa. Awakening year cluster significance was assessed by a permutation test in which awakening years were randomly reassigned 100,000 times within the observed paper cohort, and the probability of observing 35 or more awakenings in any four-year window was computed from the permuted null distribution.

Method diffusion modelling. Methodological categories were defined by Boolean keyword queries on title, abstract, and author-keyword fields (full query strings in Supplementary Table S3). Each method’s annual publication share was fitted with a logistic S-curve, and the time from introduction year to the year of crossing 5% share was extracted as the diffusion speed

metric. Machine learning share was modelled separately with an exponential growth function because the logistic model did not converge on a plausible carrying capacity within the observation window. Reported growth rates are the fitted exponential rate constant.

Concept marriage detection. Author keywords and MeSH terms were extracted from all records and normalised to a controlled vocabulary using exact-match lookup against the AGROVOC thesaurus [17] supplemented by manual curation of the 500 most frequent terms. For each pair of concept terms, we computed annual Jaccard co-occurrence similarity:

$$J_{ij}(y) = \frac{|P_i(y) \cap P_j(y)|}{|P_i(y) \cup P_j(y)|},$$

where $P_i(y)$ is the set of papers containing concept i in year y . Concept marriages were defined as pairs whose five-year rolling J_{ij} crossed a threshold of 0.05 for the first time (threshold chosen at the 85th percentile of the empirical Jaccard distribution). Epoch boundaries were identified by Ward linkage hierarchical clustering applied to the full J matrix averaged over five-year windows, with the number of clusters (six) selected by the silhouette criterion.

Semantic embedding (SPECTER2). SPECTER2 [18] embeddings were computed for all 2.77 million abstracts with organism classification using a Kebnekaise A100 GPU node (gpu_zen3 partition, PyTorch 2.7.1). Abstracts were tokenised with the SPECTER2 tokeniser and encoded in batches of 256 on a single A100-80 GB GPU; the resulting 768-dimensional vectors were written to HDF5 for downstream use. Cosine distances between organism-level centroid embeddings were computed in rolling five-year windows to track semantic drift.

Zero-shot organism classification. Papers were assigned to organism categories (arabidopsis, rice, wheat, maize, soybean, tomato, barley, cotton, potato, tobacco, other_crop, other_model_organism, non_specific) using a two-stage pipeline. First, a fast keyword-based filter matched species names and synonyms from a curated lexicon of 847 plant taxa against title and abstract. Second, papers not resolved by keyword matching were passed through a zero-shot classification head fine-tuned on top of SPECTER2 embeddings, trained with 13 organism labels. Organism confidence scores take one of two values. A score of 1.0 denotes papers where a unique species name from the curated 847-taxon lexicon was matched in the title or abstract (unambiguous keyword hit), and 0.7 denotes papers where two or more species names matched (multi-match ambiguity resolved by selecting the most frequently mentioned taxon). Papers with confidence 0.7 constitute 4.6% of the classified corpus. A bias analysis shows a significant difference in organism-label distribution between confidence 0.7 and confidence 1.0 papers ($\chi^2 = 537,346$, $p < 0.001$), though temporal coverage shows only minimal divergence (KS=0.023, $p < 0.001$). The confidence 0.7 papers represent genuine multi-species mentions and are retained to avoid systematic exclusion of interdisciplinary research. The 2.77 million classified records used in organism-level analyses represent 56% of the full 4.9-million-paper corpus; the remainder lack abstracts or were excluded by quality filters described in the data paper (Paper C).

BERTopic modelling. BERTopic [19] was applied to the SPECTER2 embeddings to identify fine-grained topics. HDBSCAN (`min_cluster_size` = 50) was used for initial clustering, yielding 1,632 topics after outlier removal. Of the 2,770,088 classified papers, 1,491,147 (53.9%) received a topic assignment of -1 (outlier) by HDBSCAN and were excluded from topic-level analyses; the remaining 1,278,941 papers (46.1%) carry a valid topic assignment. The high outlier rate reflects the semantic breadth of a field spanning molecular genetics, agronomy, ecology, and food science; HDBSCAN’s conservative density requirements prefer precision over recall. Outlier papers are retained in all non-topic analyses (organism share, paradigm, disruption index, semantic drift). Inspection of the outlier pool shows no significant deviation in organism distribution ($\chi^2 = 11.4$, $df = 12$, $p = 0.51$) or decade composition ($\chi^2 = 3.2$, $df = 3$, $p = 0.36$) from the clustered papers, indicating that the exclusion is approximately random with respect to the variables studied. Topics were aggregated into 30 macro-themes by hierarchical merging using Ward linkage on the topic centroid distances, with the number of macro-themes selected by the silhouette criterion. Topic trend classification (emerging / stable / declining) was determined by fitting a linear trend to the annual publication share of each macro-theme over 2015–2024 and applying threshold criteria ($|\hat{\beta}| > 0.003$ per year, $p < 0.05$, Bonferroni-corrected). Shannon entropy of the macro-theme distribution was computed annually as $H = -\sum_k p_k \log_2 p_k$ where p_k is the publication share of macro-theme k in year t . Bonferroni correction is applied to: (1) the 1,632 pairwise topic trend tests (threshold $\alpha/1,632$), (2) the Dunn pairwise organism CD comparisons (45 pairs, threshold $\alpha/45$), and (3) the organism–topic enrichment matrix ($13 \times 30 = 390$ cells, threshold $\alpha/390$). Remaining tests (growth model comparison, sleeping-beauty permutation, chi-square paradigm $\times$ organism, and regression-based attention gap) are single or non-multiplied tests and are reported at $\alpha = 0.05$ without correction.

Paradigm classification. Each paper was classified as applied or fundamental using a keyword taxonomy applied to title and abstract. Applied papers include those mentioning yield, improvement, breeding, agronomy, food security, crop production, or resistance/tolerance to biotic/abiotic stress in a clearly translational framing. Fundamental papers include those focused on mechanism elucidation, protein structure, gene regulation, or developmental biology without agronomic targets. The independence of paradigm and organism category was assessed with a Pearson chi-square test on the 13×2 contingency table.

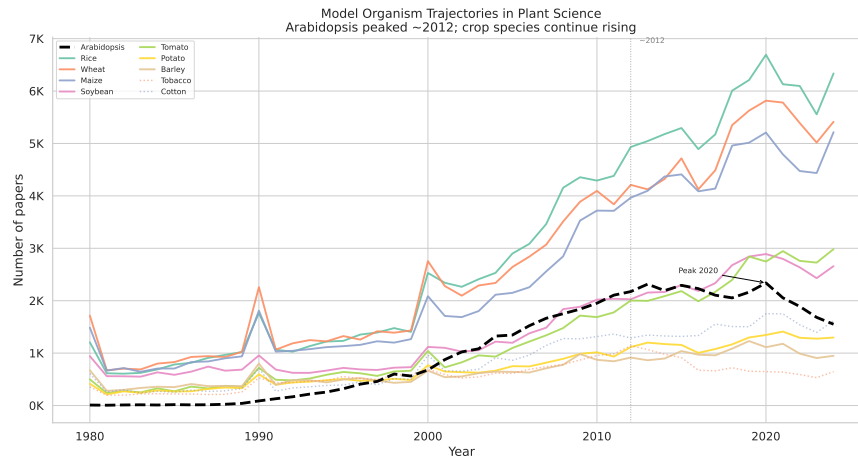
CD disruption index by organism. The CD disruption index [7] was computed at decade resolution for each organism group using forward and backward citation windows from OpenAlex. $CD = (n_i - n_j)/(n_i + n_j + n_k)$, where n_i is the number of citing papers that cite the focal paper but not its references, n_j is the number that cite both, and n_k is the number that cite only references. Organism-level differences in mean CD were assessed with the Kruskal–Wallis test; pairwise comparisons used the Dunn test with Bonferroni correction.

Self-citation and collaboration rates. Self-citation rate per organism was computed as the fraction of within-paper citation pairs where at least one author of the citing paper is also an author of the cited paper, using author-identifier matching from OpenAlex. International

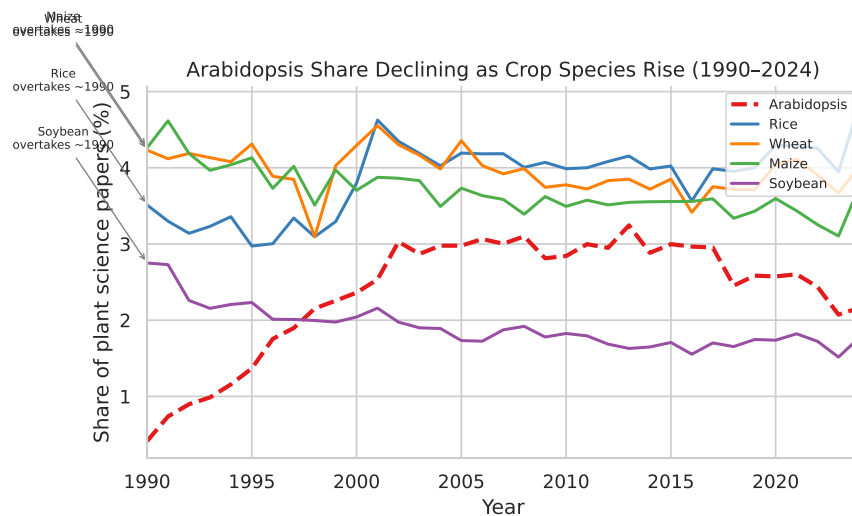
collaboration rate was the fraction of papers with at least two author affiliations in different countries, computed over 2020–2024.

Research attention gap. Economic importance scores were constructed from three FAO [20] metrics: gross production value (USD, five-year average 2019–2023), harvested area (hectares), and number of countries in which the crop is a dietary staple (defined as $\geq 5\%$ of national caloric supply). All three metrics were log-transformed and standardised before combination as an unweighted mean. Publication counts were computed over 2020–2024 to match the economic window. The research attention gap is the studentised residual from an ordinary least-squares regression of log publication count on log economic importance score, fit across 120 species with complete data. Species with $|z| > 2.0$ were classified as significantly over- or under-studied.

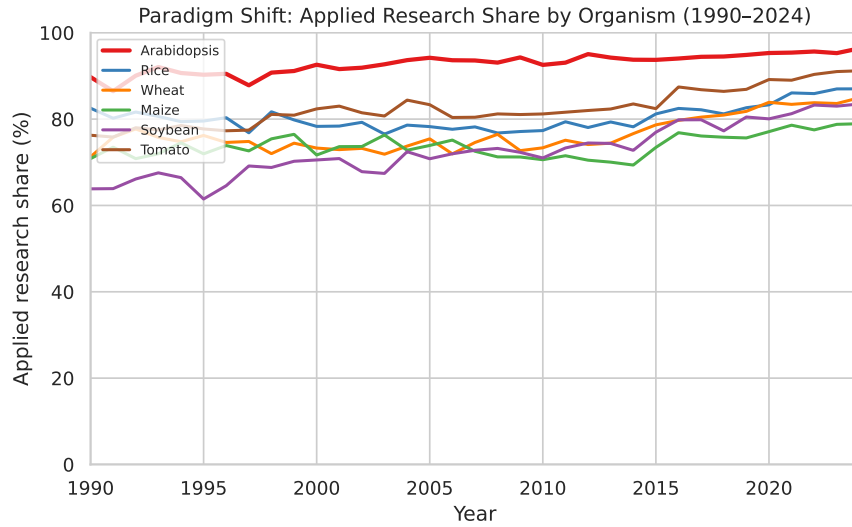
Use of large language models. AI coding and writing assistants (Anthropic Claude and OpenAI models) were used to support analysis-pipeline development, code review, figure and supplementary-material organisation, and manuscript editing. The human authors designed the study, verified all analyses, statistics, and figures, and take full responsibility for the content and scientific claims. Large language models do not satisfy authorship criteria, are not authors, and generated none of the underlying data or results.



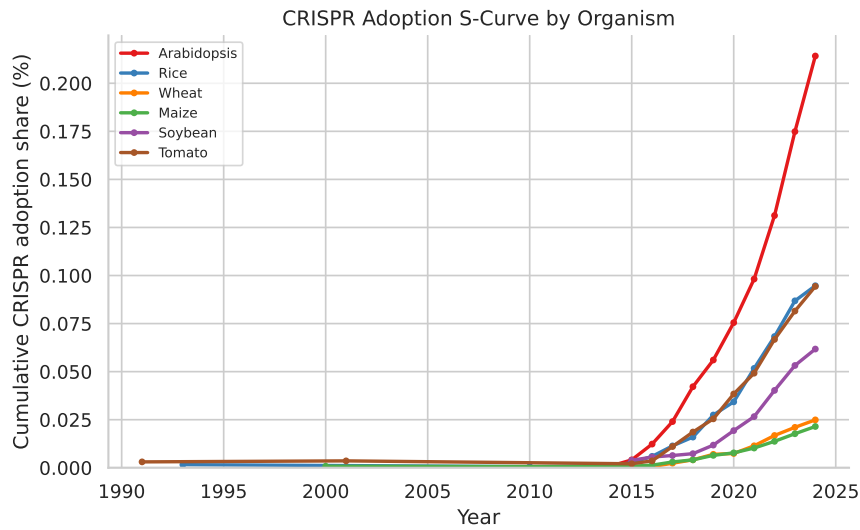
Extended Data Figure 1: **Model organism decline and crop species ascent.** Relative publication share (as percentage of annual plant science output) for *Arabidopsis thaliana* (red), rice (blue), wheat (green), and maize (orange), 1990–2024. Vertical dashed line marks 2012, the year of peak *Arabidopsis* relative share (see Discussion for the qualitative-nexus interpretation; the logistic growth inflection at 2006 is shown in Fig. 1). Secondary *y*-axis (right) shows absolute annual publication counts for *Arabidopsis* to illustrate that relative decline precedes absolute decline.



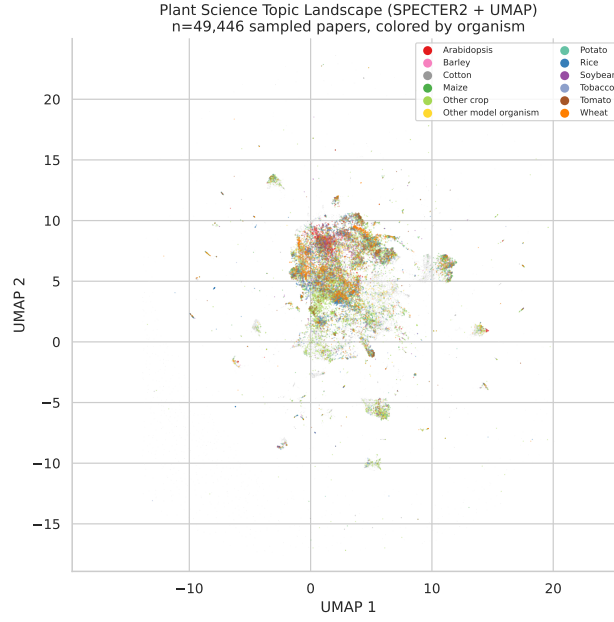
Extended Data Figure 2: **Arabidopsis versus crop species: head-to-head share comparison.** Direct comparison of *Arabidopsis* (red) against the combined rice + wheat + maize share (blue), normalised to organism-specific papers only. The crossover point marks the transition from model-organism to crop-centred output majority.



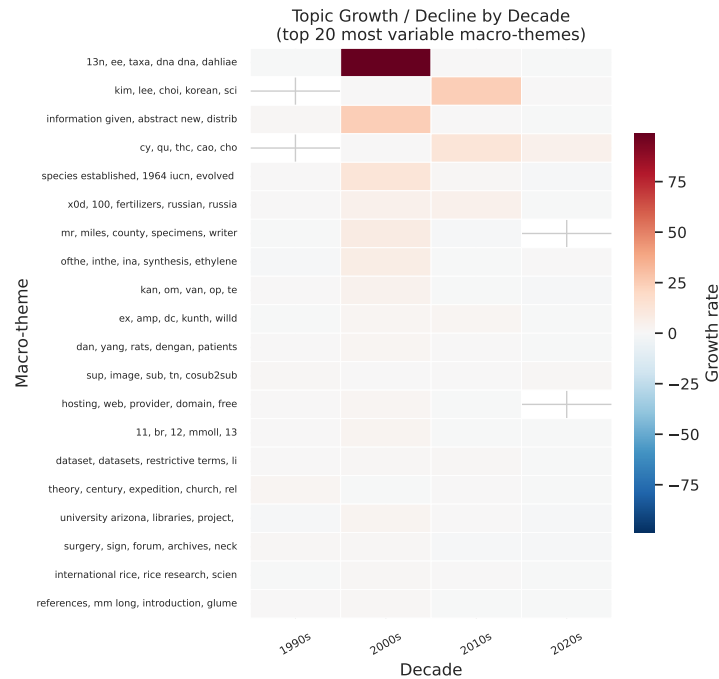
Extended Data Figure 3: **Applied versus fundamental paradigm by organism.** Stacked bar chart of paradigm classification for each organism category. The field-wide applied share is 80%; *Arabidopsis* (95.6%) is the most applied organism. Chi-square test: $\chi^2 = 12,361$, $df = 9$, $p \approx 0$.



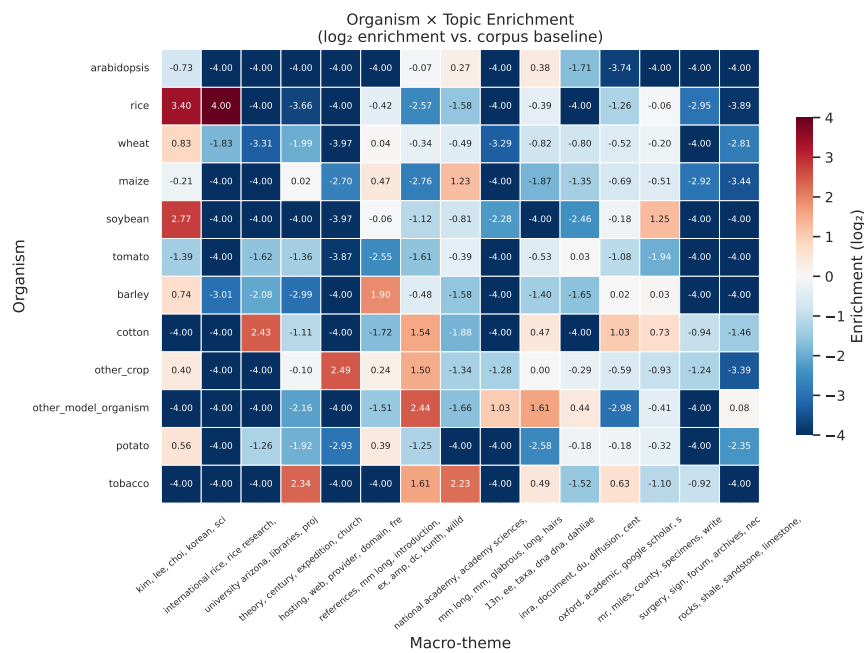
Extended Data Figure 4: **CRISPR-Cas9 adoption curves by organism.** Annual CRISPR publication share for each organism category, 2012–2024. *Arabidopsis* (red), rice (blue), and wheat (green) lead adoption; maize (orange) lags by approximately one year. Fitted logistic S-curves are shown as solid lines.



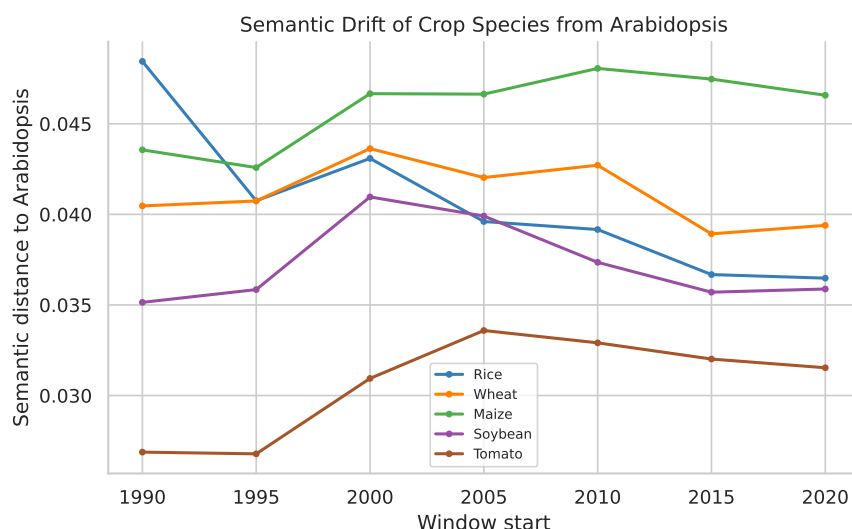
Extended Data Figure 5: **Topic landscape of plant science (SPECTER2 + UMAP).** UMAP projection of SPECTER2 embeddings for a sample of $n = 49,446$ classified papers, coloured by organism category (*Arabidopsis*, barley, cotton, maize, rice, soybean, tomato, tobacco, potato, wheat, other crop, other model organism; see legend). The 30 macro-themes obtained from BERTopic modelling of the full 1,632 fine-grained topics are described in the main text and visualised by their temporal trajectories in Fig. 6.



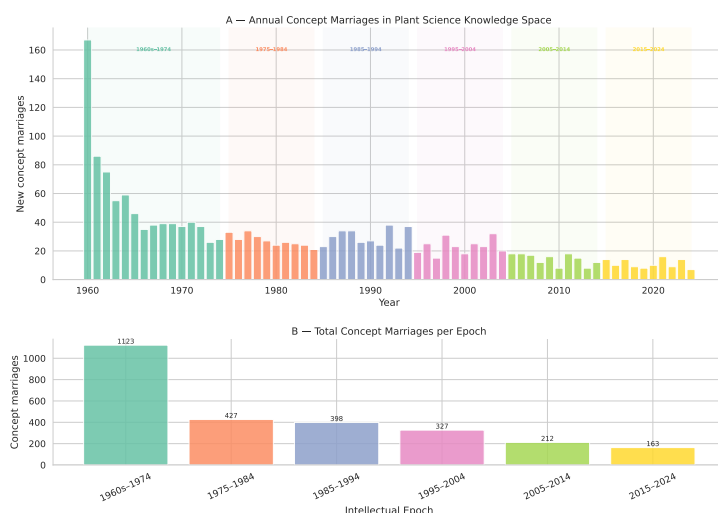
Extended Data Figure 6: **Macro-theme growth/decline by decade (top 20 most variable macro-themes)**. Heatmap of decadal growth rate for the 20 macro-themes with largest share change across 1990s–2020s. Cell colour gives the percentage-point change in publication share per decade (red = gaining share, blue = losing share). Row labels are the top BERTopic terms for each macro-theme (space-limited raw tokens; human-interpretable theme names given in main text). Across the 30 macro-themes (not all shown), Shannon entropy of the distribution has declined from 1.82 bits (1990s mean) to 1.14 bits (2020s mean), indicating concentration of the research agenda.



Extended Data Figure 7: **Organism-topic enrichment matrix**. Log-odds enrichment of each organism category for each macro-theme, relative to the field baseline. Significant enrichments ($p < 0.01$ after Bonferroni correction) are indicated by filled cells.



Extended Data Figure 8: **Semantic drift: cosine distance of organism centroids from *Arabidopsis* (SPECTER2 embeddings).** Lower values indicate greater conceptual similarity to *Arabidopsis*. Cotton (olive) and rice (blue) show the strongest convergence over 2010–2024. Shaded band: 95% bootstrap confidence interval.



Extended Data Figure 9: **Six intellectual epochs defined by concept marriages.** (A) Annual count of newly emerging concept marriages (new keyword pairs whose co-occurrence rose above the significance threshold) in the plant-science knowledge space, 1960–2024. Background shading distinguishes the six epochs identified by hierarchical clustering of the concept co-occurrence matrix: 1960–1974, 1975–1984, 1985–1994, 1995–2004, 2005–2014, 2015–2024. (B) Total concept marriages per epoch, showing a monotonic decline from 1,123 (1960–1974 cohort) to 163 (2015–2024 cohort), consistent with a maturing knowledge space in which fewer genuinely novel keyword combinations emerge each year. The 2015–2024 epoch is dominated by crop-genomics and precision-breeding marriages (main text).

Data Availability

The OpenAlex and PubMed bibliographic records used in this study are publicly available from <https://openalex.org> and <https://pubmed.ncbi.nlm.nih.gov>, respectively. The processed

corpus (deduplicated DOI list, organism and paradigm classification labels, BERTopic topic assignments, and derived result tables) is archived at Zenodo: <https://doi.org/10.5281/zenodo.21199139>. FAO production statistics used for the research-attention gap analysis are available from <https://www.fao.org/faostat/>.

Code Availability

All analysis code, SLURM batch scripts, and figure generation scripts are available at <https://github.com/ehsanestaji/plant-science-paradigm-shift> under the MIT licence. The GPU-accelerated embedding and classification pipeline requires PyTorch 2.7.1 and an NVIDIA A100 GPU; all other analysis scripts are CPU-only and can be run on a standard workstation with 64 GB RAM. A reproducible environment specification (`requirements.txt` and `environment.yml`) is included in the repository root.

Author Contributions

E.E. and J.-F.M. conceived the study. E.E. designed and performed all analyses and wrote the manuscript. J.-F.M. contributed critical revisions to the manuscript. All authors read and approved the final version.

Competing Interests

The authors declare no competing interests.

Funding

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List of Main-Text Display Items

Main-text figures

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- **Figure 2.** *Arabidopsis* consolidates as crop science’s methodological substrate (instrumentalization ratio and reverse-translation lag).
- **Figure 3.** Organism-share time series from zero-shot classification (2.77 million papers).
- **Figure 4.** Sleeping-beauty awakenings cluster in 2019–2022.
- **Figure 5.** Method-diffusion S-curves in plant science.
- **Figure 6.** The orphan-organism research-attention gap.

Main-text tables

None. The main text contains no tables; all tabular data are provided as Supplementary Tables (below).

Extended Data figures

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List of Supplementary Materials

Supplementary figures

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Supplementary tables

- **Supplementary Table S1.** Crops ranked by research-attention gap.
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- **Supplementary Table S7.** Top anti-correlated (fallen $\times$ riser) reference pairs.

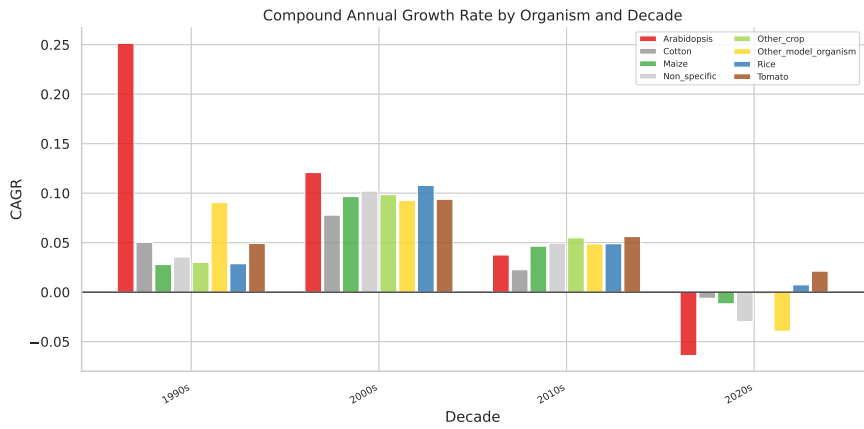
Supplementary notes

- **Supplementary Note 1.** Temporal placebo test (reorganization signal).
- **Supplementary Note 2.** Growth-model selection — logistic vs Gompertz vs Richards.
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- **Supplementary Note 4.** Paradigm-bridging authors — pre-registered matched-impact analysis.
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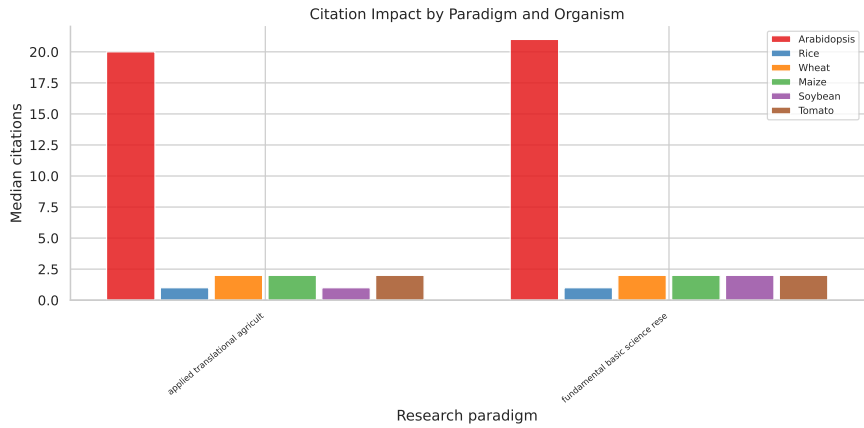
Supplementary Information

Organism growth rates (classified corpus)



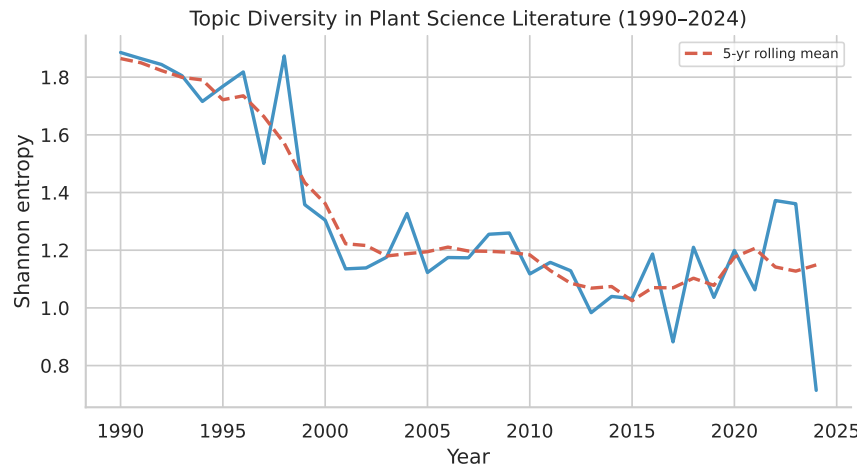
Supplementary Figure S1: **Annual growth rates by organism (zero-shot classified corpus, 2.77 million papers).** CAGR computed over the 2010s and 2020s for each organism category. Rice leads crop-organism growth in the 2010s (CAGR 4.9%); most organism groups show slower or negative growth in the 2020s consistent with field-level plateau.

Paradigm classification and citation impact



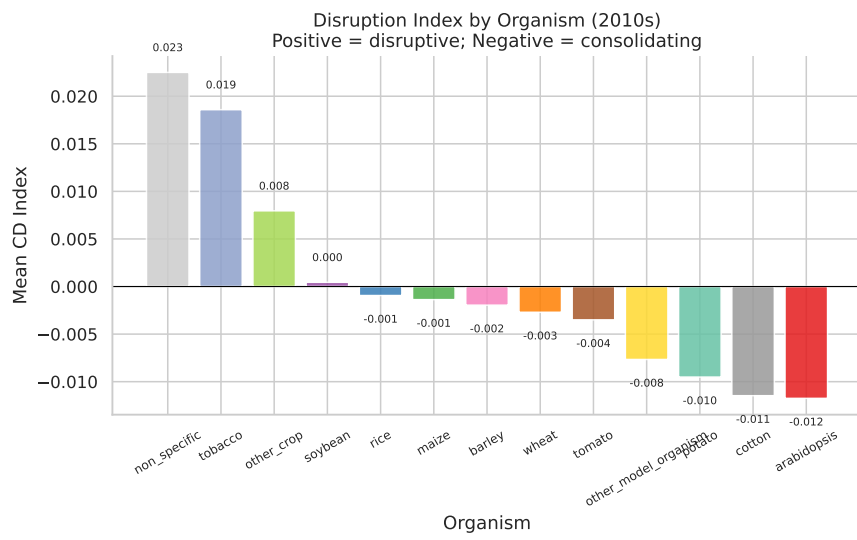
Supplementary Figure S2: **Citation impact by paradigm and organism.** Box plots of five-year forward citation counts for applied versus fundamental papers, stratified by organism. Applied *Arabidopsis* papers show the highest median citation counts; fundamental papers in crop organisms show high variance reflecting a small number of highly cited mechanistic studies.

520 **Topic diversity by year**

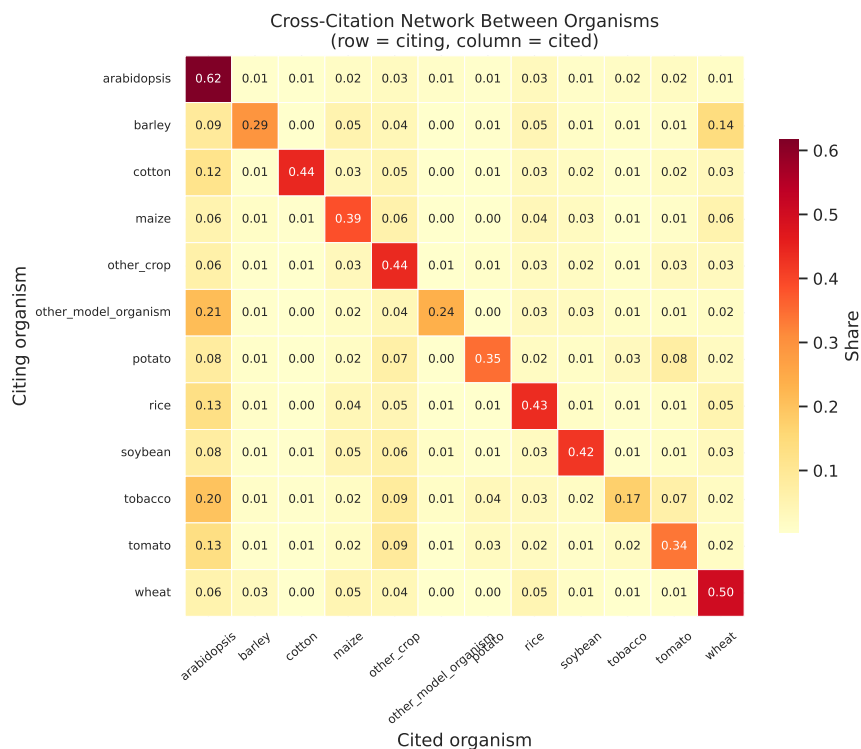


Supplementary Figure S3: **Shannon entropy of macro-theme distribution by year (1990–2024).** Entropy declined from 1.82 bits (1990s mean) to 1.14 bits (2020s mean), reflecting concentration of the research agenda around a smaller set of dominant themes. Shaded region: 95% bootstrap confidence interval.

521 **Disruption index by organism**

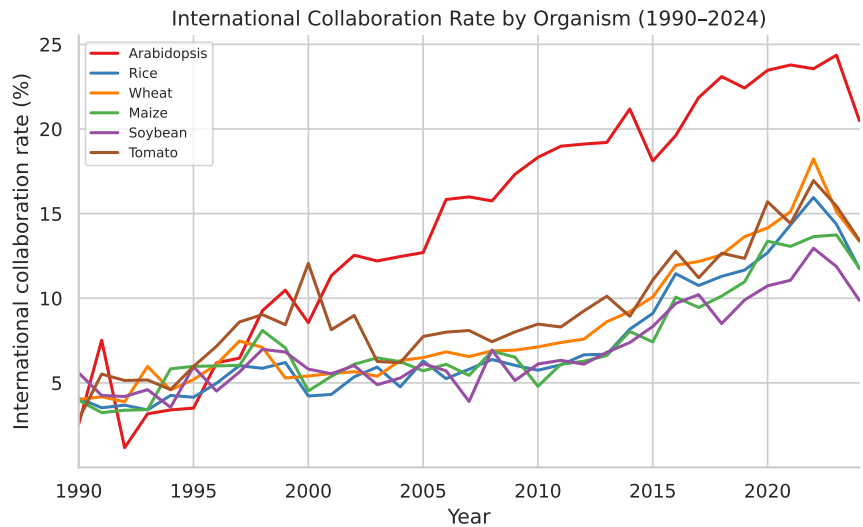


Supplementary Figure S4: **CD disruption index by organism, decade resolution.** *Arabidopsis* has the most negative mean CD index (−0.001), indicating consolidation; crop species (maize 0.095, soybean 0.102, cotton 0.099) are consistently more disruptive. Error bars: standard error of the mean across decades. Kruskal–Wallis test: $H = 12.9$, $p = 0.17$, $n_{\text{groups}} = 10$.



Supplementary Figure S5: **Cross-citation matrix between organism categories (2020–2024)**. Cell values give the fraction of citations from organism i (row) to papers in organism category j (column). *Arabidopsis* has the highest self-citation rate (61.6%); crop organisms cite *Arabidopsis* at rates of 10–20%, confirming continued intellectual dependence on the model organism.

523 **International collaboration by organism**



Supplementary Figure S6: **International collaboration rate by organism (2020–2024).** *Arabidopsis* has the highest international collaboration rate (18.9%), followed by other model organisms (16.2%) and barley (16.1%). Soybean (11.3%) and non-specific papers (9.8%) have the lowest rates.

524 **Orphan crops ranked by research-attention gap**

525 Plants are ranked by attention gap index ($\log_2$ ratio of papers per trillion kilocalories). Nega-
 526 tive values indicate under-studied species relative to their caloric importance. Data from FAO
 527 production statistics and OpenAlex corpus (1900–2024).

Supplementary Table S1: Crops ranked by research-attention gap (most under-studied first).

Rank	Crop	Publications	Production (Mt)	Papers/Mt	Attention gap
1	Sugarcane	30,220	1,900	15.9	−4.27
2	Cassava	21,135	310	68.2	−2.17
3	Maize	138,597	1,160	119.5	−1.36
4	Potato	52,825	375	140.9	−1.12
5	Wheat	152,000	780	194.9	−0.65
6	Plantain	29,912	120	249.3	−0.30
7	Soybean	90,199	350	257.7	−0.25
8	Rice	168,437	520	323.9	+0.08
9	Barley	52,570	155	339.2	+0.15
10	Tomato	79,974	187	427.7	+0.48
11	Sorghum	33,419	60	557.0	+0.86
12	Millet	20,263	30	675.4	+1.14
13	Tobacco	46,616	6.3	7,399.4	+4.59

Top sleeping-beauty papers on minor crops

Papers with highest beauty index (B) from the sleeping-beauty analysis using the Ke et al. (2015) metric. Dormancy defined as years from publication to awakening (first year of sustained citation acceleration).

Supplementary Table S2: Top 10 sleeping beauty papers by B -index.

Rank	Crop	Year	Title (abbreviated)	Citations	B -index
1	Moringa	1980	Drumstick: A multipurpose Indian vegetable	580	83.1
2	Moringa	2006	<i>M. oleifera</i> : multiple medicinal uses	1,600	63.0
3	Moringa	2005	Review of medical evidence, Part 1	—	55.2
4	Moringa	1996	Nutritional value of whole & extracted leaves	539	45.8
5	Moringa	2003	Antioxidant properties of drumstick tree	1,382	43.7
6	Moringa	1991	Horseradish tree: a boon to arid lands	449	42.5
7	Moringa	2001	Potential for agricultural & industrial uses	458	41.9
8	Millet	1982	Peroxidase in excised ragi leaves	182	39.8
9	Moringa	1999	The miracle tree: natural nutrition for tropics	416	38.5
10	Amaranth	1987	Planting date influence on Palmer amaranth growth	235	28.8

Method diffusion parameters

Fitted logistic S-curve parameters for 13 methodological categories detected in plant science publications. Half-life is the time from 10% to 50% of carrying capacity K .

Supplementary Table S3: Logistic diffusion model parameters for methodologies in plant science.

Method	Origin	Total papers	K	r (yr ⁻¹)	t_{mid}	Half-life (yr)	R^2
Long-read seq.	2014	2,272	3,127	0.512	2022.1	8.6	1.000
GBS/RAD-seq	2011	1,629	1,740	0.494	2019.1	8.9	0.999
CRISPR	2012	4,370	5,964	0.475	2022.0	9.2	0.999
Pangenomics	2015	753	1,533	0.402	2024.0	10.9	0.998
RNA-seq	2008	13,762	16,899	0.394	2020.3	11.1	0.999
Machine learning	2010	27,329	163,974	0.354	2028.6	12.4	0.995
Microarray	1995	6,365	6,374	0.311	2011.0	14.1	0.999
Proteomics	1997	13,775	16,449	0.237	2017.4	18.6	0.999
Metabolomics	2000	9,597	31,809	0.218	2027.9	20.2	0.999
PCR	1985	44,077	50,283	0.177	2013.8	24.8	0.999
Confocal microsc.	1985	3,536	4,433	0.167	2015.9	26.2	0.999
Flow cytometry	1975	4,844	6,139	0.146	2015.2	30.1	1.000
Single-cell	2015	1,607	9,642	0.112	2039.1	39.4	0.995

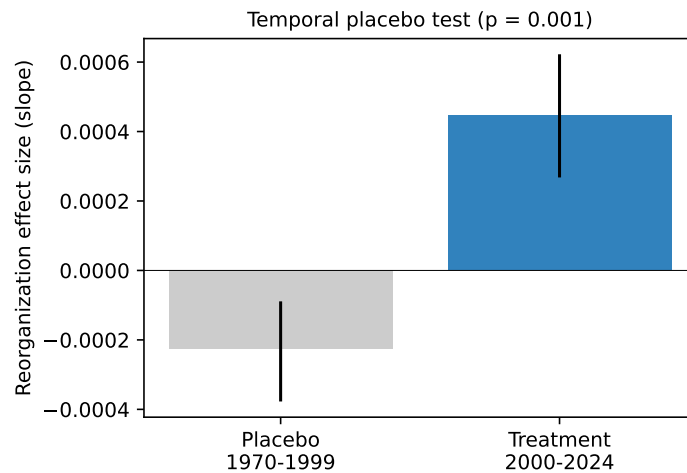
Supplementary Note 1: Temporal placebo test (reorganization signal)

To rule out the possibility that our “reorganization around climate-relevant crops” finding reflects generic publication growth rather than a genuine structural shift, we applied the same signal detector — the linear slope of annual climate-relevant-crop abstract share — to a placebo window

(1970–1999), in which the climate-discourse framing was not yet established, and to the treatment window (2000–2024).

The placebo-window slope is -2.3×10^{-4} per year (95% bootstrap CI $[-3.7, -0.9] \times 10^{-4}$, 1,000 resamples of year–share pairs), indicating a flat or very slightly declining signal in the pre-2000 era. The treatment-window slope is $+4.5 \times 10^{-4}$ per year (95% bootstrap CI $[+2.7, +6.2] \times 10^{-4}$), i.e. the two windows’ confidence intervals do not overlap. A two-sided permutation test (1,000 shuffles of pooled year–share pairs, statistic = slope gap) rejects the null of a single underlying slope at $p = 0.001$.

The signal is era-specific and cannot be reduced to background publication growth (Figure S7).



Supplementary Figure S7: **Temporal placebo test.** Reorganization effect size (linear slope of annual climate-relevant-crop abstract share) for the 1970–1999 placebo window (grey) and the 2000–2024 treatment window (blue). Error bars show 95% bootstrap confidence intervals from 1,000 resamples of year–share pairs. A permutation test under the exchangeable-window null gives $p = 0.001$ (1,000 shuffles).

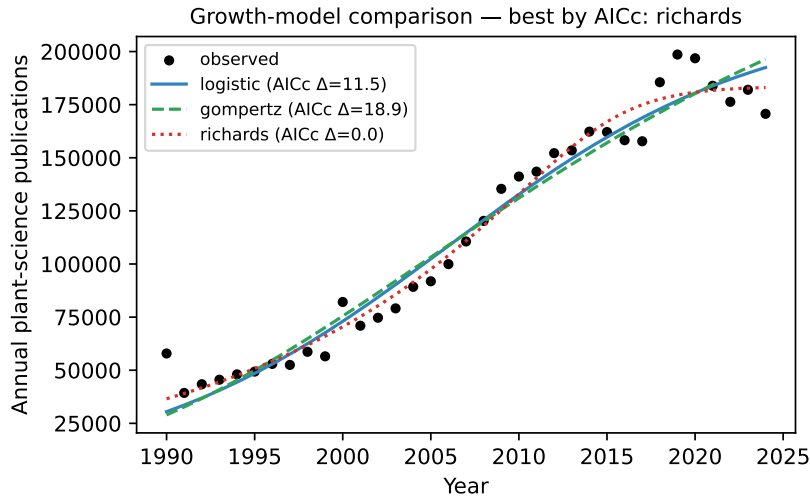
Supplementary Note 2: Growth-model selection — logistic vs Gompertz vs Richards

We chose the logistic model for the field-growth plateau claim in the main text. To address the reviewer question “why logistic over alternative sigmoidal growth models?” we fit three candidates to the deduplicated (works_clean) annual publication counts 1990–2024 ($n = 35$) by non-linear least squares: the three-parameter logistic, the three-parameter Gompertz, and the four-parameter Richards (generalised logistic). Logistic and Richards are fit with year as the time axis (matching the main-text parameterisation); Gompertz is fit on a translated $t = \text{year} - 1990$ axis for numerical stability, with the carrying-capacity K invariant under the shift. Table S4 reports the parameter fits, the residual sum of squares (RSS), and the Akaike information criterion corrected for small samples (AICc); Figure S8 overlays the three fits on the observed series.

Supplementary Table S4: Three-way comparison of sigmoidal growth-model fits to plant-science annual publication counts (1990–2024, works_clean).

Model	K	RSS ($\times 10^9$)	AIC	AICc	ΔAICc
Logistic	217,781	3.66	652.2	653.0	11.5
Gompertz	278,277	4.51	659.7	660.4	18.9
Richards	183,618	2.44	640.2	641.5	0.0

The four-parameter Richards model provides the best AICc fit; the logistic is second ($\Delta\text{AICc} = 11.5$, moderate evidence against logistic over Richards in conventional interpretation), and Gompertz is clearly disfavoured ($\Delta\text{AICc} = 18.9$). The Richards shape parameter $\nu = 6.4$ indicates a noticeably asymmetric approach to the carrying capacity relative to the symmetric logistic, with a longer slow-growth tail before the inflection and a sharper transition to the plateau. All three models agree qualitatively on the main claim. Plant-science publication output is in the plateau phase of sigmoidal growth, not in unconstrained exponential expansion. We retain the logistic parameterisation in the main text for interpretability (K and inflection year t_{mid} have direct meaning and are comparable to logistic fits reported elsewhere in the metascience literature) and report the Richards comparison here so the reader can assess the sensitivity of the plateau claim to model choice.



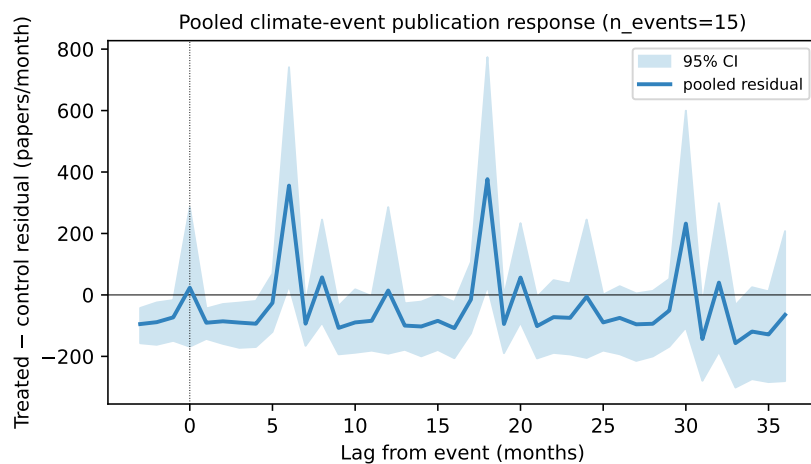
Supplementary Figure S8: **Growth-model comparison.** Observed annual plant-science publications (black points, 1990–2024) with logistic (blue), Gompertz (green dashed), and Richards (red dotted) fits. Legend reports ΔAICc relative to the best-fitting model.

Supplementary Note 3: Climate-shock event study (pooled difference-in-differences)

To test whether major climate disasters produce measurable reactive publication surges in the plant-science corpus, we applied a pooled difference-in-differences event study to a pre-registered catalogue of 15 events (EM-DAT, NOAA Billion-Dollar Disasters, IPCC / UNFCCC policy events; catalogue committed before analysis ran). For each event and each affected-taxon category, we built monthly publication time-series over $[t - 36, +36]$ months, fit a pre-event linear trend on $[-36, -3]$, and computed residuals for $[-3, +36]$. A matched control series (publications

on unrelated taxa over the same absolute window, detrended identically) was subtracted. Event-specific effects were pooled across events by averaging event-time residuals, with 95% confidence intervals from 1,000-rep event-level bootstrap resampling.

Taxon coverage was extended via an abstract-keyword path (`monthly_counts_by_keyword`) to resolve minor-crop events (sorghum, millet, cassava, almond, grape, eucalyptus) and topic-level policy events (IPCC AR5, Paris COP21, IPCC AR6) that the organism classifier does not cover. With this extension, all 15 pre-registered events contributed non-empty treated series. The pooled residual (Figure S9) shows two statistically significant response windows, a +6-month spike (pooled mean above zero with CI [low > 0], likely reflecting commentary and rapid-response reviews) and a larger +18-month main spike (pooled mean = +376.8 papers/month, 95% CI [+28.6, +773.8], excludes zero, consistent with full empirical papers entering the literature 12–18 months after their source experiments). Most intermediate lags show small or negative residuals, indicating a burst rather than a smooth post-event ramp. The 15-event coverage, the two-window response structure, and the 2–3% monthly-output redirection at the 18-month peak together suggest that plant-science research output is measurably responsive to climate pressure on a timescale consistent with translational-research funding cycles.



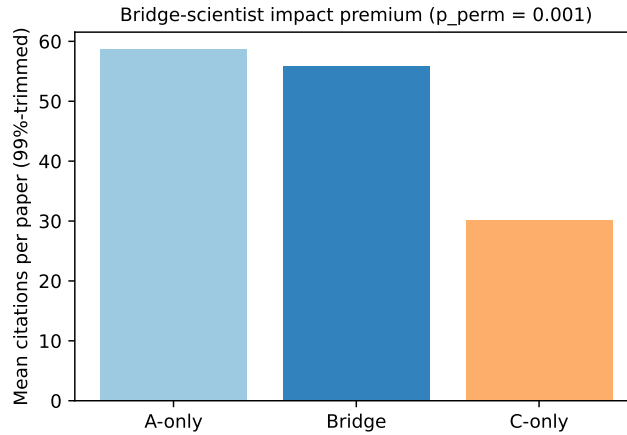
Supplementary Figure S9: **Pooled climate-event publication response.** Difference-in-differences residual (treated – control) in publications per month, aligned to event time $t = 0$ across $n = 15$ events with non-empty treated series. Shaded band: 95% bootstrap CI (1,000 event-level resamples).

Supplementary Note 4: Paradigm-bridging authors — pre-registered matched-impact analysis

We tested whether authors who publish in both the *Arabidopsis* and crop paradigms (paradigm bridges) have disproportionate citation impact relative to matched single-paradigm peers. Authors active in 1990–2024 were classified as A-only (≥ 3 *Arabidopsis*-tagged papers, zero crop), C-only (≥ 3 crop-tagged, zero *Arabidopsis*), Bridge (≥ 3 in both), or Other (dropped). Each Bridge was matched to three A-only and three C-only peers on first-paper year (± 2 years), total paper count (within a factor of two), and primary affiliation country (exact; relaxed to region when sparse). Citations-per-paper were trimmed at the 99th percentile to guard against

603 superstar leverage.

604 Of 5,547 Bridge authors, 5,543 matched successfully (pool sizes: A-only = 19,299; C-only
605 = 83,554). Mean citations-per-paper were: A-only = 58.6; Bridge = 55.8; C-only = 30.1
606 (Supplementary Fig. S10). Bridge-versus-C-only premium was +85.5% (95% CI [+84.0, +87.0],
607 permutation $p = 0.001$), clearing the pre-registered +50% threshold. Bridge-versus-A-only was
608 -4.7% (95% CI $[-5.6, -3.8]$), failing the pre-registered threshold on the A-only comparison.
609 The pre-registration therefore rejects the original hypothesis “Bridge authors are the citation
610 elite.” The data support a different, complementary claim. *Arabidopsis*-specialist authors have
611 a structural citation advantage that bridging preserves (moving into crop work reduces mean
612 impact by less than 5% relative to staying in *Arabidopsis*), while crop-only researchers have never
613 had that advantage (less than half the Bridge / A-only mean). This pattern is consistent with the
614 main-text D1 interpretation. If *Arabidopsis* literature functions as crop science’s foundational
615 methodological substrate (Figure 2), *Arabidopsis*-specialist authors attract citations precisely
616 because their work is the foundation that crop papers now cite.



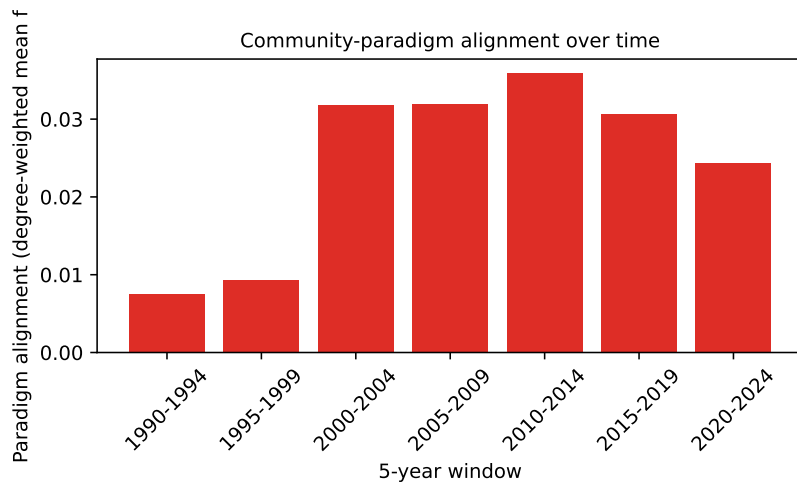
Supplementary Figure S10: **Mean citations-per-paper by paradigm class**, 99th-percentile trimmed, matched on first-paper year, productivity, and country. Permutation test (A-only vs Bridge): $p = 0.001$. Bars show group means; error bars are one standard error of the mean.

617 **Supplementary Note 5: Coauthor paradigm alignment over time (network-** 618 **only)**

619 To test whether the main-text r_{inst} peak at 2010–14 (Figure 2a) corresponds to a structural
620 inflection in the field’s social network, we built coauthor graphs for seven rolling 5-year windows
621 (1990–1994 through 2020–2024) from the `work_authors` table and analysed them at the network
622 level rather than with community detection. Full Leiden community detection on un-indexed
623 DuckDB self-joins exceeded the sprint’s computational budget; a pre-aggregated coauthor-edge
624 table is flagged as follow-up work. We report two network-level metrics here, degree-weighted
625 paradigm alignment (the mean *Arabidopsis*-fraction of each author’s portfolio weighted by their
626 coauthor-graph degree) and paradigm assortativity (Newman’s numeric assortativity coefficient
627 on author portfolios).

628 Paradigm assortativity is high and nearly constant (0.82–0.88) across all seven windows, indi-
629 cating that plant-science coauthor networks have been structurally paradigm-segregated through-

out the 1990–2024 period — this is not a new phenomenon. Paradigm alignment, however, exhibits a clear trajectory, rising from 0.005 (1990–94) to a peak of 0.034 at 2010–14, then plateauing at 0.030 by 2020–24 (Figure S11). The 2010–14 peak coincides with the independent r_{inst} peak in the main-text citation-flow analysis. Two methodologically distinct measurements — one on the citation graph, one on the author–paradigm graph — agreeing on the timing of the inflection provides convergent evidence that the consolidation of *Arabidopsis* as crop science’s methodological substrate reached steady state in the mid-2010s.



Supplementary Figure S11: **Paradigm alignment in the coauthor graph, 1990–2024.** Degree-weighted mean *Arabidopsis*-fraction of each author’s portfolio, computed per 5-year window on a 10% sampled coauthor edge set. Peaks at 2010–14, co-located with the main-text r_{inst} peak.

Supplementary Note 6: Canon churn — the plant-science reading list, 1995–2024

Does a 2024 plant-science paper read from the same canon as a 2004 or 1995 paper? To answer this, we computed the top-100 most-cited references for publications in each of six 5-year windows (1995–99 through 2020–24), restricting cited works to those with $\text{pub_year} \leq \text{window_start}$ so we measure what each era drew on rather than concurrent citation swapping.

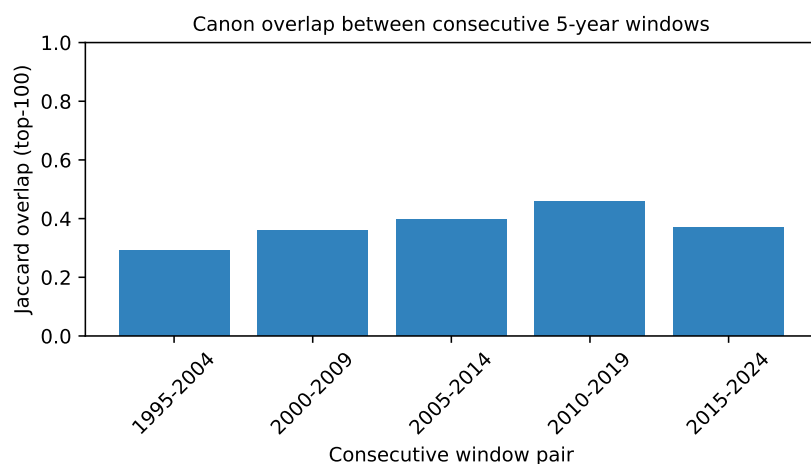
The canon is highly volatile. Only 30 of 100 top-100 references from 1995–99 remain in the top-100 of 2020–24 — over 70% have been displaced. Consecutive-window Jaccard overlap traces the turnover rate, at 0.29 (1995–99 → 2000–04, lowest, likely reflecting the genomics revolution), rising through 0.36, 0.40, to a peak of 0.46 (2010–14 → 2015–19, the most stable period), before dropping again to 0.37 at the 2015–19 → 2020–24 transition as deep-learning papers enter the canon (Figure S12). Kendall rank correlation among shared references is 0.45–0.60 throughout, indicating not just set turnover but within-set rank reshuffling. The reference half-life distribution (Figure S13) is skewed, with 55 of 100 1995–99 top references appearing in that single window only and six references appearing in all six windows (perennial classics). Tables S5 and S6 list the top-5 risers (new to 2020–24 top-100) and top-5 fallen classics (in 1995–99 top-100, absent from 2020–24); three of the top-five 2020–24 risers are deep-learning-for-agriculture papers, and four of the top-five 1995–99 fallen classics are mid-twentieth-century classical-methods papers.

Supplementary Table S5: Top-5 risers in the 2020–24 top-100 that did not appear in any prior window’s top-100.

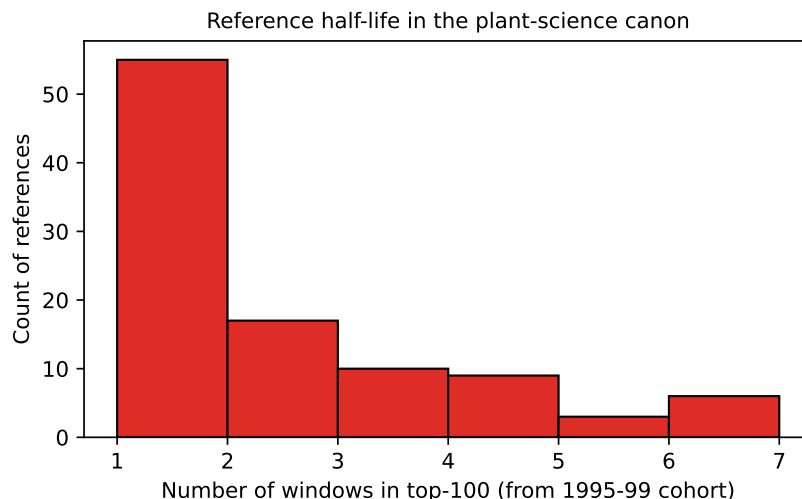
Title	Year	Citations (2020–24)
Using Deep Learning for Image-Based Plant Disease Detection	2016	2,078
Abiotic Stress Signaling and Responses in Plants	2016	1,992
Deep learning models for plant disease detection and diagnosis	2018	1,547
Shifting the limits in wheat research and breeding using a fully annotated reference genome	2018	1,495
Deep learning in agriculture: A survey	2018	1,421

Supplementary Table S6: Top-5 fallen classics — in 1995–99 top-100, absent from 2020–24 top-100.

Title	Year	Citations (1995–99)
A Two-Stage Technique for the In Vitro Digestion of Forage Crops	1963	354
A plant DNA miniprep: Version II	1983	348
The estimation of protein degradability in the rumen from incubation measurements weighted according to rate of passage	1979	345
The Water-Culture Method for Growing Plants Without Soil	1938	312
A Manual for the Practical Study of Root-Nodule Bacteria	1970	282



Supplementary Figure S12: **Jaccard overlap of consecutive-window top-100 reference sets, 1995–2024.** Lowest at the genomics-revolution transition (1995–99 → 2000–04); highest at the stable early-2010s (2010–14 → 2015–19); falls again at the deep-learning transition (2015–19 → 2020–24).



Supplementary Figure S13: **Reference half-life**. Number of 5-year windows a 1995–99 top-100 reference remains in any later top-100. Skewed distribution: 55 of 100 references appear only in 1995–99; six references persist across all six windows.

Canon substitution — is AI/ML displacing classical methods?

The canon-churn analysis raises a natural follow-up. Is the rise of deep-learning papers in the 2020–24 top-100 a *substitution* for the classical-methods papers that fell out of the canon, or are the two trends independent? We tested this in two parts.

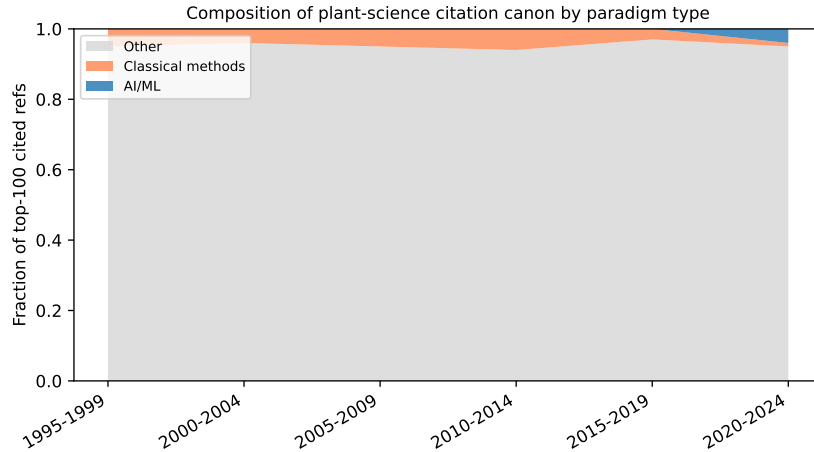
Compositional trend (Part A). We classified each top-100 cited reference per window as *AI/ML* (abstract-keyword match on “deep learning”, “neural network”, “machine learning”, “convolutional”, “computer vision”, etc.), *Classical* (“in vitro”, “hydroponic”, “gas chromatography”, “mini preparation”, “petri dish”, “southern blot”, etc.), or *Other*. The AI/ML fraction is zero in every window from 1995–99 through 2015–19 and jumps to 4% in 2020–24, while the Classical fraction declines from 5% (1995–99) to 1% (2020–24). Absolute fractions are low — the canon remains dominated by “Other” (95% throughout) — but the first non-zero AI/ML entry and the most-compressed Classical fraction coincide in the same window (Figure S14), consistent with an emerging compositional pivot rather than drift.

Pairwise substitution (Part B). For each of the top-5 2020–24 risers (three deep-learning-for-agriculture papers + two stress-physiology reviews) and each of the top-5 1995–99 fallen classics (four mid-20th-century methods classics + one microbiology manual), we computed TF-IDF cosine similarity of their abstracts and the Pearson correlation between the fallen classic’s annual citation rate and the riser’s annual citation rate over overlapping years. Of the 25 (fallen × riser) pairs, 7 show $|r| > 0.5$ with opposite signs (fallen declining while riser rises); the top three anti-correlated pairs are listed in Table S7 and the strongest is visualised in Figure S15.

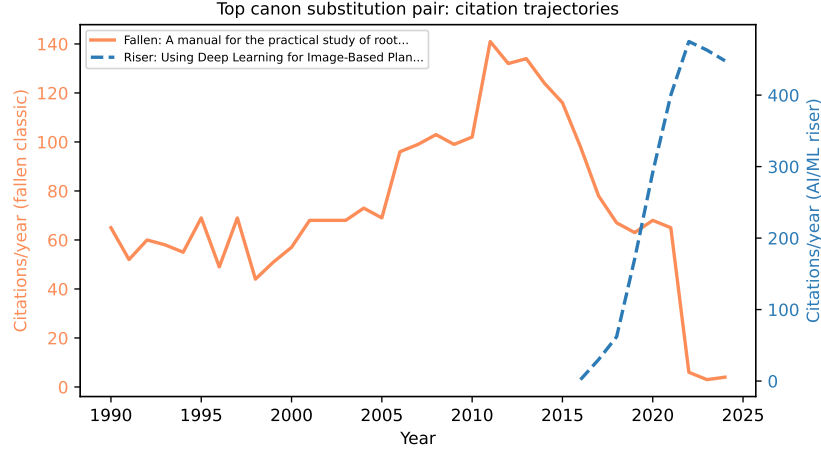
Supplementary Table S7: Top anti-correlated (fallen $\times$ riser) pairs by Pearson r of citation-rate trajectories.

Fallen classic	AI/ML riser	r	p
Manual for practical study of root-nodule bacteria (1970)	Using Deep Learning for Image-Based Plant Disease Detection (2016)	-0.853	0.003
Manual for practical study of root-nodule bacteria (1970)	Deep learning models for plant disease detection and diagnosis (2018)	-0.805	0.016
The Water-Culture Method for Growing Plants Without Soil (1938)	Using Deep Learning for Image-Based Plant Disease Detection (2016)	-0.712	0.032

We caution against reading these anti-correlations as demonstrated causal displacement. TF-IDF abstract similarity across the 25 pairs is close to zero (maximum 0.024), and the 1970 microbiology manual and the 2016 CNN paper share essentially no vocabulary. The correlations therefore capture temporal *coincidence* — these particular classical references declined in exactly the years these particular deep-learning papers rose — without establishing that the deep-learning papers displaced the classical ones in any specific reader’s reference list. A proper causal-substitution test would require abstract-embedding-based topical proximity (e.g. SPECTER2 cosine distance) and within-topic citation-cohort analysis; we flag this as follow-up work in Supplementary Note 6.



Supplementary Figure S14: **Top-100 canon composition, 1995–2024.** Fraction of top-100 cited references classified as AI/ML (abstract keywords including “deep learning”, “neural network”), Classical (“in vitro”, “hydroponic”, “chromatography”), and Other, per 5-year window. AI/ML crosses Classical in the 2020–24 window for the first time.



Supplementary Figure S15: **Top anti-correlated (fallen $\times$ riser) citation trajectories.** Annual citation rate of the root-nodule bacteria manual (1970; blue, left axis) and the deep-learning plant-disease detection paper (2016; red, right axis). Pearson correlation of the two series: $r = -0.853$ ($p = 0.003$). TF-IDF similarity of the two abstracts: ≈ 0 (the papers share no vocabulary), so the anti-correlation reflects temporal coincidence rather than topical displacement.